Jae Jung Kim

+82-10-9595-3492 | rlawownd0905@unist.ac.kr | Google Scholar

Education

Ulsan National Institute of Science and Technology (UNIST) *Feb 2023 – Present*
Combined M.S./Ph.D. Program in Mechanical Engineering; Advisor: Prof. Chun Sang Yoo

Ulsan National Institute of Science and Technology (UNIST) *Feb 2023*
B.S. in Mechanical Engineering; Minor in Physics; Summa Cum Laude (GPA: 3.99/4.3)

- National Scholarship for Science and Engineering

Kyeongbuk Science High School, Pohang, Republic of Korea *Feb 2019*
Graduated early (two-year program)

Research Experience

Clean Combustion & Energy Research Laboratory, UNIST *Feb 2023 – Present*
Graduate Researcher (Combined M.S./Ph.D. Program)

- Conducting high-fidelity combustion simulations and developing reduced-order models for turbulent reacting flows.
- Completed alternative military service obligation during the Ph.D. program (Republic of Korea).

Clean Combustion & Energy Research Laboratory, UNIST *Dec 2020 – Feb 2023*
Undergraduate Research Intern

Journal Publications

1. **Kim, J.J.**, Lacey, C.E., Lee, J.H., Jung, K.S., Chen, J.H., and Yoo, C.S., "Time-dependent-bases with local CUR decomposition method for accelerating turbulent combustion simulations," *Combustion and Flame* 286, 114848 (2026). DOI: 10.1016/j.combustflame.2026.114848.

2. Oh, S.Y.*, **Kim, J.J.***, Jung, K.S., and Yoo, C.S., "A DNS study of differential diffusion effects on the ignition of a turbulent $NH_3/H_2/N_2$-air mixing layer," *Combustion and Flame* 288, 114936 (2026). DOI: 10.1016/j.combustflame.2026.114936.

* Equal contribution (co-first authors).

Conference Presentations

1. "Accelerating Operator-Splitting Reacting-Flow Simulations Using the TDB-CUR Algorithm." *64th Symposium (Japanese) on Combustion*, Fukuoka, Japan, Nov 4–6, 2026.
2. "Acceleration of stiff chemistry calculations in operator-splitting-based reacting flow simulations using the TDB-CUR algorithm." *2026 Spring KSME Symposium*, Ulsan, Republic of Korea, May 22, 2026.
3. "Development of an OpenFOAM-based library for acceleration of stiff chemistry via TDB-CUR algorithm." *71st KOSCO Symposium*, Busan, Republic of Korea, May 13–15, 2026.
4. "Time-Dependent-Bases with Local CUR Decomposition Method for Reduced-Order Modeling and Stiffness Mitigation in Combustion Simulations." *69th KOSCO Symposium*, Samcheok, Republic of Korea, May 2025.
5. "Time-Dependent-Bases with Local CUR Decomposition Method for Turbulent Combustion Simulations." *16th International Conference on Combustion and Energy Utilization*, Hong Kong, Feb 2025.
6. "Numerical Study on the Flame Structures and Edge Dynamics of DME Nonpremixed Counterflow Cool Flame." *66th KOSCO Symposium*, Jeju, Republic of Korea, Nov 2023.

Selected Research

OpenFOAM-Based Acceleration of Operator-Splitting Chemistry *May 2025 – Present*
- Developed an OpenFOAM-based TDB-L-CUR framework for operator-splitting reacting-flow solvers, targeting expensive stiff chemistry integration.

Local Low-Rank Modeling and Stiffness Mitigation for Combustion *Aug 2024 – Dec 2025*
- Developed the TDB-L-CUR approach combining on-the-fly local manifold construction, adaptive sampling, and computational singular perturbation (CSP) analysis.

Differential Diffusion in Turbulent Ammonia–Hydrogen Ignition *2026*
- Investigated ignition dynamics in turbulent $NH_3/H_2/N_2$-air mixing layers using DNS, with a focus on the competing effects of differential diffusion and turbulent mixing.

Chemical Explosive Mode Analysis (CEMA) for OpenFOAM *Mar – Apr 2025*
- Implemented CEMA diagnostics in OpenFOAM-6 and identified and corrected an issue in the chemistry Jacobian implementation.

Earlier Fundamental Combustion Research *2020 – 2024*
- DME counterflow diffusion cool flame and flame-edge dynamics.
- Premixed spherical-flame instabilities
- Ammonia/*n*-heptane RCCI engine combustion

Industrial Projects

Liquid Ammonia Spray | POSCO *Jul 2025 – Nov 2026*
- Ignition and combustion characteristics of liquid ammonia sprays

Industrial Burner | POSCO *Apr 2024 – Mar 2025*
- Combustion of hydrocarbon/hydrogen/ammonia blends in industrial burners

Gas Turbine | KEPCO Research Institute *Apr 2023 – Mar 2024*
- Combustion of methane/ammonia blends in a full-scale can-type gas turbine

Rotary Kiln Burner | POSCO *Jan 2023 – Jul 2023*
- Combustion simulations of a 40-m-scale industrial rotary kiln burner

Gas Oven | LG Electronics *Jun 2021 – Jan 2023*
- Combustion analysis for CO emission reduction in household gas ovens

Technical Skills

Programming: C++, Fortran, Python, MATLAB
CFD: OpenFOAM (v6/v12), ANSYS Fluent, COMSOL
Languages: Korean (native), English (fluent)